

YOLO Based Eye Tracking System for Real-Time CVI

Team C10

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1st Review Presentation

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Introduction

- Cerebral Visual Impairment (CVI) is a condition where the brain has difficulty in processing visual information
- People with CVI may experience difficulty in focusing, tracking, and interpreting visual objects.
- Regular visual activities can help improve visual attention and eye brain coordination.
- Traditional eye tracking systems can be expensive and require specialized hardware.
- Our project proposes a low-cost gaze-controlled game using a webcam.

Problem Statement

- To develop a low-cost, webcam-based gaze-controlled game that supports visual training for individuals with CVI, a condition with no direct cure

Objectives

- To develop a low-cost gaze-controlled game using a standard webcam.
- To convert gaze movement into game controls for interactive gameplay.
- To provide an engaging visual exercise that encourages eye–brain coordination.
- To create an accessible alternative to expensive eye-tracking hardware.

Literature Review

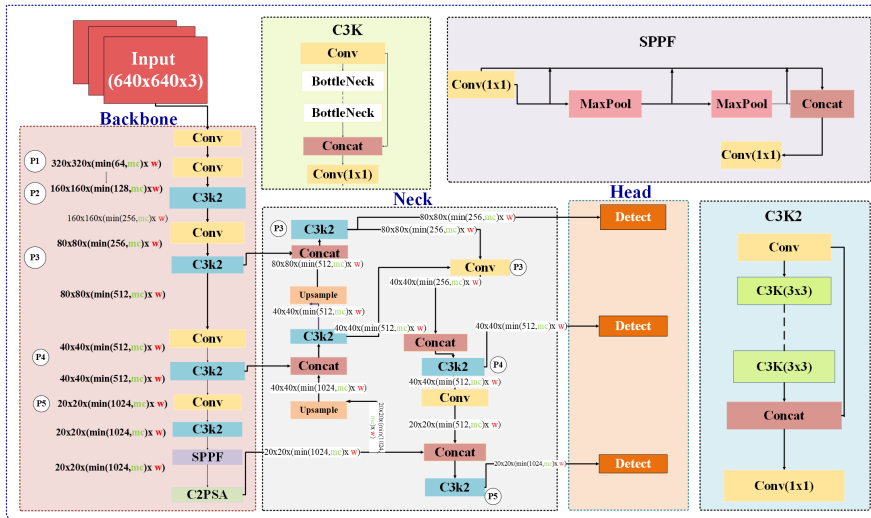
S.No	Paper / Venue	Technique	Inference	Limitations
1	Serious Games to Support Cognitive Development in Children with Cerebral Visual Impairment	Mobile games, Eye Tracker, Touch Interface	Serious games improve attention and rehabilitation	Eye tracker is only used for assessment. Does not estimate precise iris coordinates or provide gaze controlled gameplay.
2	Robust Pupil Detection for Real-Time Pervasive Eye Tracking	Canny Edge Detection, Morphological Processing, Ellipse Fitting, Edge Selection	Achieved 72.02 pupil detection accuracy within 5 pixels and real-time operation at 120 FPS.	Sensitive to occlusions, reflections, blur, and weak edges.
3	Automatic Pupil Detection and Gaze Estimation Using the Vestibulo-Ocular Reflex in a Low-Cost Eye-Tracking Setup	Python, OpenCV, Webcam, Pupil Detection, VOR, Projection Mapping	Low-cost real-time pupil detection.	Requires head mounted hardware and calibration;

Literature Review

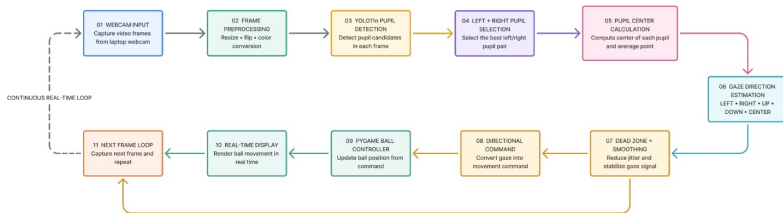
S.No	Paper / Venue	Technique	Inference	Limitations
4	Gaze Detection – A Better Way to Type	OpenCV, Dlib, 68 Facial Landmarks, Webcam, Blink Detection, Virtual Keyboard, Python/NumPy, Computer Vision	Hands-free typing using gaze and eye blinks with a virtual keyboard.	Gaze accuracy, blink detection, and reliability need improvement.
5	Deep Learning Models for Webcam Eye Tracking in Online Experiments	Webcam, OpenCV, Deep Learning, MPIIGaze, ETHXGaze, FAZE	FAZE achieved 2.44° gaze accuracy and 0.47° precision using webcam-based tracking.	Lower accuracy than specialized eye trackers and high computational requirements for real-time tracking
6	Deep-Learning-Based Pupil Center Detection and Tracking Technology for Visible-Light Wearable Gaze Tracking Device	YOLOv3-tiny, visible-light camera	Real-time pupil tracking with less than 5 px error and 20 FPS.	Fixed head position; needs head-movement compensation

- Existing systems often use expensive eye-tracking hardware.
- Limited research focuses on webcam-based iris tracking for CVI.
- Lack of engaging game-based activities that encourage repeated visual practice.

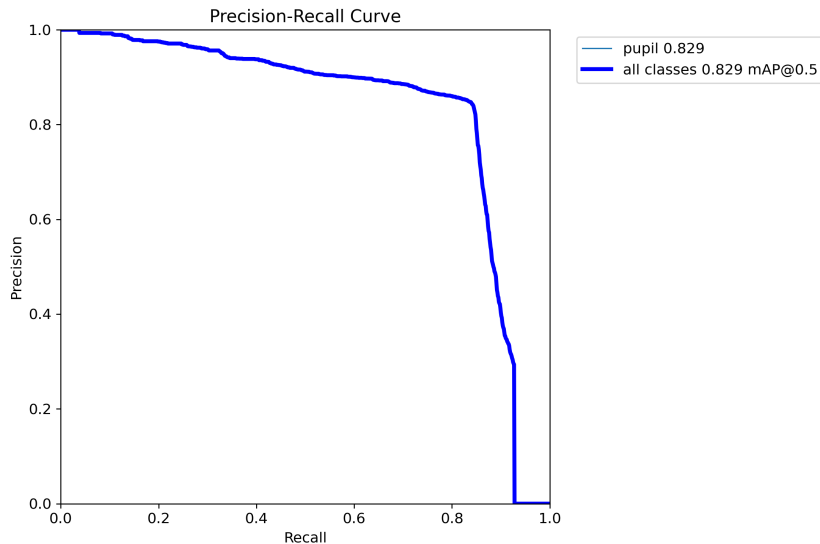
YOLO11n Architecture



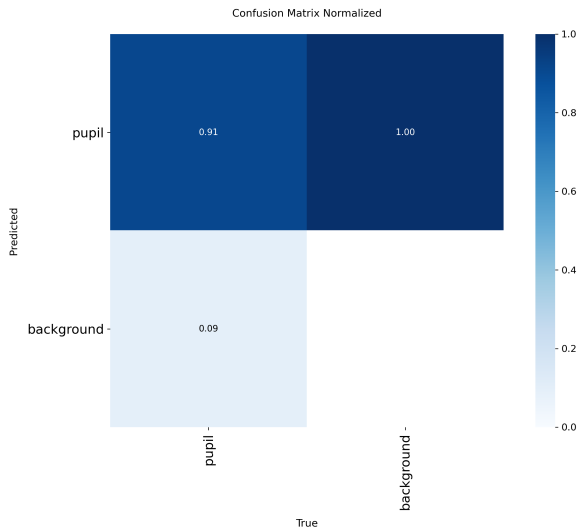
REAL-TIME APPLICATION PIPELINE (After Running the Trained YOLO Model)



Results



Results



Results

